

# Priyanshu Arora

Python Backend Engineer | AI Engineer

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## Professional Summary

Python Backend Engineer and AI Engineer with **2+ years of experience** building cloud-native backend services and AI-powered applications using **Python, FastAPI, AWS, Docker, and REST APIs**. Experienced in developing solutions for the **US healthcare domain**, with hands-on experience handling **HIPAA-sensitive data, PII, secure data exchange, and healthcare data pipelines**. Skilled in **serverless architectures, infrastructure as code, performance optimization**, and production AI systems involving **LLMs, RAG pipelines, AI agents, and prompt engineering**. **Certified Ethical Hacker (CEH)** with a strong foundation in backend security, cloud deployment, databases, and secure application development.

## Technical Skills

**Languages:** Python, SQL, JavaScript, TypeScript

**Backend:** FastAPI, Django, Django REST Framework, Flask, REST APIs, SQLAlchemy, Pydantic

**Cloud & DevOps:** AWS, AWS Lambda, API Gateway, EC2, S3, RDS, DynamoDB, SQS, CloudWatch, AWS CDK, Docker, CI/CD, Linux

**AI / Generative AI:** LLMs, Generative AI, RAG, AI Agents, Prompt Engineering, NLP, OpenAI APIs, Vector Search, Embeddings

**Databases & Testing:** PostgreSQL, MySQL, pgvector, pytest, unittest, Postman

**Tools:** Git, JIRA, GitHub, Nginx

## Professional Experience

### Spark Eighteen Pvt. Ltd.

Backend Engineer | Cloud & AI Systems

Udaipur, Rajasthan, India

Apr 2024 – Present

- Built secure **PII tokenization pipelines** for US healthcare data using **Python and AWS services**, enabling protected data exchange while strengthening data privacy and secure handling of confidential information.
- Architected a **zero-downtime data migration pipeline** using staged migration, automated validation, and controlled cutover, enabling migration from a legacy data source to an automated system while **reducing infrastructure costs** and eliminating cross-account operational overhead.
- Designed and deployed **cloud-native backend services** on AWS using Lambda, API Gateway, EC2, S3, and RDS, improving scalability, reliability, automation, and cost efficiency across backend workloads.
- Defined and managed **AWS infrastructure as code using AWS CDK**, automating resource provisioning, deployment workflows, and repeatable environment setup.
- Improved **AWS Lambda execution performance by 10x** through database restructuring, workflow optimization, and code-level performance tuning, significantly reducing latency and operational overhead.
- Developed and maintained **Python/FastAPI backend APIs and data workflows** for cloud-native applications, integrating authentication, database operations, and AWS services into production workflows.

### GKM IT Pvt. Ltd.

Python Backend Intern

Udaipur, Rajasthan, India

Sept 2023 – Mar 2024

- Developed and maintained **Python backend APIs using Django REST Framework and Flask**, implementing models, views, serializers, validation, and RESTful API workflows for production applications.
- Worked across the backend development lifecycle including **API development, database integration, debugging, authentication, authorization, and API testing** using Python, SQL, and Postman.
- Applied secure development practices and authentication concepts while completing **Certified Ethical Hacker (CEH)** and Google Cybersecurity certifications, strengthening practical knowledge of application and API security.

### Cnetric Global Inc.

Software Engineer Intern

Remote

Mar 2023 – Jun 2023

- Developed backend integrations for a customized e-commerce PaaS solution using **Shopify GraphQL APIs and Node-RED automation workflows**, supporting data exchange and workflow automation.

## Education

### Indian Institute of Technology (IIT) Ropar

Major in AI, GPA: 7.8/10

Ropar, India

Sept 2024 – Feb 2026

### Techno India NJR Institute of Technology

Bachelor of Technology in Computer Science, GPA: 9.7/10

Udaipur, Rajasthan, India

Sept 2020 – May 2024

## Certificates And Achievements

Certified Ethical Hacker (CEH)

IBM Cloud Computing

IBM Data Analysis using Python

30 Days of Postman

Google Cybersecurity

Microsoft: Python Programming

IBM Data Visualization using Python

Upgrad: Database Design